

Question Number	Answer	Additional guidance	Mark
8(a)(i)	An explanation that makes reference to two of the following points: • {cats / rats} {ate / destroyed / preyed on} the {eggs / chicks / plovers} (1) • few(er) chicks to {reproduce / survive to breeding age} / fewer birds to reproduce (next year) (1) • increased competition for food / {rats / cats} carried diseases which killed the birds (1)	ACCEPT fewer chicks hatch ACCEPT death rate higher than birth rate IGNORE habitat loss	(2)

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	36	REJECT answers given as fractions	(1)

Question Number	Answer	Additional guidance	Mark
8(b)(i)	An answer that makes reference to two of the following points: • consideration of analysis of {alleles / gene pool} (1) • take pollen from one gumwood tree and use it to pollinate another gumwood tree (1) • planting of {seeds / young saplings / shoots} {where they won't be grazed by rabbits / in a safe area / protected area / area with no rabbits} (1)	IGNORE store in seedbanks ACCEPT {maintenance / increase} of genetic {variation / diversity} of trees ACCEPT outbreeding / breed trees with different alleles REJECT if context of a different organism ACCEPT {artificial / manual} {pollination / fertilisation} ACCEPT descriptions of {artificial / manual} {pollination / fertilisation} ACCEPT {asexual reproduction / cloning} of St Helena trees IGNORE captive breeding / IVF IGNORE planting unqualified IGNORE captivity IGNORE comments relating to {planting / protecting} {adult / existing} trees	(2)

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	An answer that makes reference to the following points: • build {fences / barriers} around young trees / spray trees with chemical to deter rabbits (1) • to prevent rabbits from being able to eat {trees / shoots} (1) Or • {kill rabbits / reduce rabbit population / introduce predators} (1) • to reduce risk of young trees being eaten / fewer {trees / shoots} are eaten (1)	ACCEPT grow in areas with no rabbits e.g. greenhouse IGNORE give rabbits other sources of food	(2)

Question Number	Answer	Additional guidance	Mark
8(b)(iii)	A description that includes four of the following points: • St Helena gumwood seeds would be selected from {different trees / different areas / genetically different trees} (1) • seeds would be x-rayed (1) • seeds would be prepared by being {dried / dehydrated} (1) • seeds would be treated with an antimicrobial (1) • seeds would be stored in suitable conditions (1)	ACCEPT collect seeds with different alleles REJECT collect seeds from different species ACCEPT {germination / growth / pollination} of plants to collect new seeds ACCEPT store in {dry conditions / low humidity} ACCEPT sterilised / disinfected IGNORE wash seeds / clean seeds IGNORE sterile container e.g. seeds are frozen / placed in {temperatures below 0°C / freezer} / store in a vacuum IGNORE low temperatures / cool space	(4)

Question Number	Answer	Additional guidance	Mark
8(c)(i)	An answer that includes the following points: • $N(N-1)$ correctly calculated (1) • $\sum n(n-1)$ correctly calculated (1) • $D=1.44$ (1)	818120 566906 ecf for mp3 ACCEPT standard form to 2 d.p. correct answer on answer line scores 3 marks	(3)

Question Number	Answer	Additional guidance	Mark
8(c)(ii)	• {agree / yes} because the {D value / diversity index / number} for {habitat 1 / 1.44} is smaller / agree because the {D value / diversity index / number} for {habitat 2 / 2.80} is larger	ecf applies ACCEPT agree because habitat 2 has more species / converse IGNORE habitat 2 has higher biodiversity	(1)